
PERGAMON

Journal of the Mechanics and Physics of Solids
49 (2001) 2095–2111

JOURNAL OF THE
MECHANICS AND
PHYSICS OF SOLIDS

www.elsevier.com/locate/jmps

The atomic dynamics of fracture

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Abstract

A personal account of "breaking" nanocrystals on the computer is presented. © 2001 Published by Elsevier Science Ltd.

Keywords: Crack; A. Fracture; Atoms; A. Dynamics; Failure

1. Introduction

Consider a smooth, straight crack in a perfect solid. Let's stretch the solid perpendicular to the direction that the crack is pointing. If the solid material is "*brittle*" and we stretch the solid enough, the crack will start moving with an increasing speed. Theory tells us that if the crack maintains its motion in the original forward direction (that is to say, it remains straight!), then the crack will eventually reach a constant velocity equal to the Rayleigh speed; that's the speed of sound on a solid surface. Why the Rayleigh speed? It's because elastic energy given to the solid by the stretch must be transferred to the crack tip in order to break atomic bonds for crack extension, and this can go only as fast as the speed of sound on the crack's surface. That appears to be a simple and reasonable description of "*how fast can cracks propagate*?" However, that's not how cracks behave in most cases.

I became interested in "*how fast cracks can propagate*" because cracks don't travel straight and they don't go as fast as the Rayleigh sound speed, or that's what the experimenters told me. In actual fact, there is no such thing as a perfect solid in the real world, and maybe this is the reason for the discrepancy. We know that real solids have all sorts of imperfections such as grain boundaries, impurities, and other cracks! Maybe this is the reason for crooked cracks and slower speeds. Actually, this is not the only reason, and that makes the story really interesting.

My tools are the molecular dynamics simulation method coupled with the use of big computers. Similar findings have been simulated at the mesoscopic scale using

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0022-5096/01/$ - see front matter © 2001 Published by Elsevier Science Ltd.
PII: S0022-5096(01)00028-X

the finite-element method within the framework of the cohesive-surface formalism (Needleman, 1987, 1999).

2. Studying materials failure by computer experiments

With the present-day supercomputers, simulation is becoming a very powerful tool for providing immediate insights into the nature of fracture dynamics. Atomistic simulations yield "*abinitio*" information about crack tip formation and deformation at length scales unattainable by experimental measurement and unpredictable by continuum elasticity theory and, hence, gives additional insights into the complex mechanisms of materials failure. It becomes possible to perform an absolutely clean experiment of fracture with defect-free solids, and we are coming up with some exciting results. Like the laboratory experiments, our initial interest was to study dynamic brittle failure. We were able to follow the crack propagation over sufficient time and distance intervals so that a comparison with experiment became feasible. A detailed comparison between laboratory and computer experiments have shown commonality in behavior at the macroscopic scale. Using our "*computational microscope*", we can see what is happening at the atomic scale.

Our simulation tool is computational molecular dynamics (Abraham, 1986), and it is very easy to describe. Molecular dynamics predicts the motion of a given number of atoms governed by their mutual interatomic interaction, and it requires the numerical integration of the equations of motion, "*force equals mass times acceleration or* $F = ma$". We learn in beginning physics that the dynamics of two atoms can be solved exactly. Beyond two atoms, this is impossible except for a few very special cases, and we must resort to numerical methods. A simulation study is defined by a model created to incorporate the important features of the physical system of interest. These features may be external forces, initial conditions, boundary conditions, and the choice of the interatomic force law. My choice of simple interatomic force laws is dictated by my intent to investigate the *generic* features of a particular many-body problem common to a large class of real physical systems and not governed by the particular complexities of a unique molecular interaction. It is very important to emphasize this conscious choice since it is not uncommon to hear others object since I am not studying "real" materials.

The simple interatomic potential may be thought of as a "*model potential*", and my model potential is normally the Lennard-Jones 12 : 6 potential. My "*model material*" is typically a two-dimensional atomic crystal but not always. Gordon (1984) comments on Griffith's desire to have a simpler experimental material that would have an uncomplicated brittle fracture. He writes, "In those days, models were all very well in the wind tunnel for aerodynamic experiments but, damn it, who ever heard of a model material?"

When I started out in the mid 1960s, a few hundred atoms could be treated. In 1984, we reached 100,000 atoms. Before that time, computational scientists were concerned that the speed of scientific computers could not go much beyond four Gigaflops, or four billion arithmetic operations per second and that this plateau would be reached by the year 2000! That is history. With the concept of concurrent computing, a modern parallel computer is made up of several (tens, hundreds or thousands) small computers working simultaneously on different portions of the same problem and sharing information by

"*communicating*" with one another. The communication is done through "*message passing*" procedures. The present record is well over a few Teraflops for optimized performance, and we recently simulated 1,000,000,000 atoms in a fracture study at the San Diego Supercomputer Center. Moore's Law states that the speeds of computers double every 1 and $1\frac{1}{2}$ years. For 35 years, that translates into a computer speed increase of ten million. This is exactly the increase in the number of atoms that I could simulate over the last 35 years.

Why do we refer to "*experiments*" in the title of this section? In a certain sense, one can view the simulation methodology as giving experimental information from a model, idealized universe. This approach has a mixture of theory and experiment, and it has been labeled "*the third kind of science*". By computationally solving the theoretical models in detail, a wealth of information about a physical phenomenon is produced that can be viewed as experimental data. Like any other experimentalist, we have to ask the right questions and have to measure the right things.

3. The experimental results: a dynamic instability, transonic & supersonic crack motion and a dynamic brittle-to-ductile transition

3.1. The dynamic instability

Continuum fracture theory typically assumes that cracks are smooth. For dynamic cracks, it predicts that they accelerate to a limiting speed equal to the Rayleigh speed of the material. However, experiment tells us that in a common fracture sequence in brittle materials, an initially smooth and mirror-like fracturing face begins to appear misty beyond a speed of about one-third the Rayleigh speed and then evolves into a rough, hackeled region as the crack accelerates to a limiting velocity of about six-tenths the Rayleigh speed. The origin of the disagreement between theory and experiment suggests questions of microscopic impurities or imperfections in the material, experimental uncertainties, or failure of theory. All of these features are unexplained using continuum theory, and experiment suggesting a crack instability beyond one-third of the Rayleigh wave speed perked my interest in dynamic brittle fracture (Fineberg et al., 1991, 1992).

In order to get started, I decided to simulate the brittle fracture of one of the simplest atomic system conceivable: a perfect two-dimensional (2D) solid composed of a sheet of atoms with their motion restricted to the plane. Why two-dimensions? First, it is less demanding computationally than three-dimensions. Second, for small deformations the triangular lattice is elastically isotropic, and third, the two-dimensional lattice of atoms interacting with a model potential was believed to be brittle. My chosen system size was as large as two million atoms, lots of atoms in 1993. Why two million atoms? A 2D wafer of one million atoms has one thousand atoms on a side, which I found large enough to simulate the early stages of fracture dynamics. The slab length was long enough for the crack to reach its maximum speed before the elastic stress waves produced by the advancing fracture reflected off the wafer's edges and return to the fracture point. Size demands more atoms, and more atoms demand more computing,

not only because of the increased size but also because the natural time scale expands; that is, it takes more time for a crack to traverse the larger system.

The atomic slab was initialized at a very low temperature, and a notch of specified size was cut midway on a slab side. An outward strain rate was imposed on the outer most columns of atoms defining the opposing perpendicular sides of the slab. This means that we are pulling on the opposing sides parallel to the notch so that the slab is being stretched a constant percentage per unit time. An increasing lateral strain occurs in the solid slab as time progresses. At a sufficiently large strain, depending upon the choice of notch length and other things, the crack begins to move from the notch tip, and the simulation is continued until the growing crack has traversed the total length of the slab. What did we expect to see? Our special solid has perfect atomic packing with no crystal imperfections except for the created notch or crack. We expected to see perfect crack motion predicted by theory; that is, with sufficient strain, the crack would accelerate to the Rayleigh wave speed with mirror surfaces being created as it passed through the solid. We were surprized!

Fig. 1 summarizes the important findings of our typical 2D fracture simulation. We see that the brittle crack initially propagates in a straight line and leaves mirror-like cleaved surfaces. However, at the crack speed of one-third of the Rayleigh wave speed (second image), the crack begins to roughen, and then to oscillate back and forth (third image) achieving a forward speed equal to approximately two-third of the Rayleigh speed. Similarly, for speeds less than one-third of the Rayleigh speed, the acceleration of the crack tip is quite smooth; however, the instantaneous tip speed becomes very erratic after reaching one-third of the Rayleigh speed. All of these features are in agreement with the laboratory findings but not with our expectation that theory would describe our idealized experiment. We didn't do just one simulation. We did many. We changed the crack length; cracks with a longer length begin moving at a lower strain and accelerate slower. We found the same thing. We changed the interatomic potential (or how the atoms interact); this changes the type of material and its sound speeds. We found the same thing. We tried bigger atomic slab sizes, different strain rates, and lots of other things. We found the same thing. We concluded that this transition from smooth, mirror-like motion to rough, erratic motion is an instability in the dynamics of the crack. It has nothing to do with material imperfections; there weren't any in our perfect solid. It has nothing to do with a particular material; we tried different interatomic potentials. *It has to do with the crack dynamics. The dynamics has an instability in the straight-line motion at one-third of the Rayleigh wave speed*! This is a very important finding.

Dislocations appear after the onset of the instability and are apparent in the gray-scale pictures. They appear as dark vertical dashes in the late-time snapshot. By examination, we observe that the transverse dislocations go out some distance, then return to the fracture surface where they disappear. The spacings between these dislocations are quite regular. We can understand their origin as a transverse slippage between two neighboring rows of atoms arising from a growing shear stress at the crack tip as the ever-increasing cascade of broken bonds in the forward direction allows ever increasing severed rows of atoms to want to relax laterally. This buildup of severed atomic rows will eventually be sufficiently large enough to overcome the barrier to slippage. The

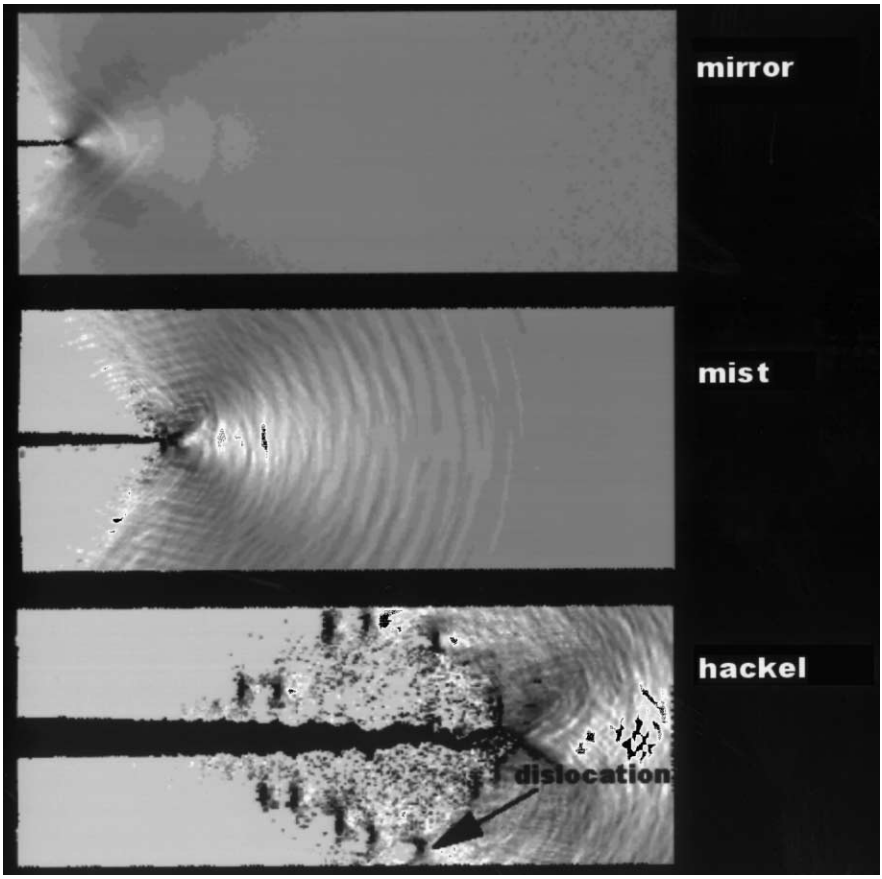


Fig. 1. The dynamic regimes of brittle fracture. The brittle crack initially propagates in a straight line and leaves mirror cleaved surfaces. At the crack speed of one-third of the Rayleigh wave speed (the second image), the crack begins to roughen, and then to oscillate back and forth (the third image) achieving a forward speed equal to two-thirds of the Rayleigh speed.

front to slippage will manifest itself as a dislocation. Of course, this is a repeating process, hence there is a continual creation of transverse dislocations. Also, the return of dislocations to the crack surface and the healing of the surface is a consequence of neighboring bands of matter bounded by slip planes relaxing to equilibrium. The dislocations disappear with no evidence that they exist during the fracture process. Examining the cleaved solid, one would erroneously conclude that dislocations are not present in the brittle fracture of our two-dimensional solid.

From a simple geometrical consideration, the oscillating zigzag motion of the crack tip explains why the "*apparent forward*" crack speed is slower than theoretical prediction. It is not quite complete, but it gets it down to approximately eight-tenth of the Rayleigh speed. There are delays in the zigzagging of the crack that must be accounted for and which further reduces the speed estimated by simple geometry. If we

could prevent the crack from zigzagging, its forward motion would be much faster and maybe equal to the Rayleigh speed. The atomic simulations are discussed in Abraham et al. (1994, 1997a, b).

3.2. Transonic crack motion

So far, I have discussed mode I (tensile) loading of a simple two-dimensional crystal with all of the atoms interacting with the same model interatomic potential. However, there are other ways to break a crystal, and the crystal structure may not be so ideal. For example, there is mode II shear loading. First, lets consider what theory tells us about the different loading.

It is widely believed that a brittle crack cannot propagate faster than the Rayleigh wave speed. The origin for this belief stems from the predictions of continuum mechanics; in particular, the dynamic elastic solutions of the near-tip stress fields and energy release rates for various velocity regimes and different types of external loading (Freund, 1990; Broberg, 1999). For a mode I (tensile) crack, both the energy release rate and the stress singularity vanish for all crack velocities in excess of the Rayleigh wave speed. The implication is that there is no possibility for a mode I crack to propagate at a velocity greater than the Rayleigh wave speed. A mode II (shear) crack behaves similarly to a mode I crack in the subsonic velocity range; i.e., the energy release rate monotonically decreases to zero at the Rayleigh wave speed and remains zero between the Rayleigh and shear wave speeds. However, the predictions for the two loading modes differ for crack velocities greater than the transverse wave speed. The energy release rate is zero for a mode I crack but is positive for a mode II crack over the entire range of intersonic velocities, with a maximum value at the radiation-free speed equal to the square root of two times the transverse sound speed. What has been concluded from these theoretical solutions? First, a mode I crack's limiting speed is the Rayleigh speed. Second, a mode II crack's limiting speed is also the Rayleigh speed because the "*forbidden velocity zone*" between the Rayleigh and shear wave speeds acts as an impenetrable barrier for the shear crack to go beyond the Rayleigh wave speed. The forbidden regimes for crack propagation are summarized in Fig. 2 for the different applied loads.

The first direct experimental observation of cracks faster than the transverse wave speed has been reported by Rosakis et al. (1999). They investigated shear dominated crack growth along weak planes in a brittle polyester resin under far field asymmetrical dynamic loading. The late arrival of an experiment on intersonic fracture is due, in part, to the fact that a crack in elastic homogeneous and isotropic solids has a zigzag crack path once the crack tip velocity exceeds the dynamic instability speed of one-third of the Rayleigh wave speed. Therefore, a wavy crack instability occurs at low crack velocities and prevents an exploration of the full range of possible velocities. By introducing a weak path (a layer of lower toughness), the crack growth is confined to this path.

We have performed two-dimensional molecular dynamics simulations of crack propagation along a weak interface joining two harmonic crystals (see Fig. 3). We assume that the interatomic forces are harmonic except for those pairs of atoms with a separation cutting the centerline of the simulation slab. For these pairs, the interatomic

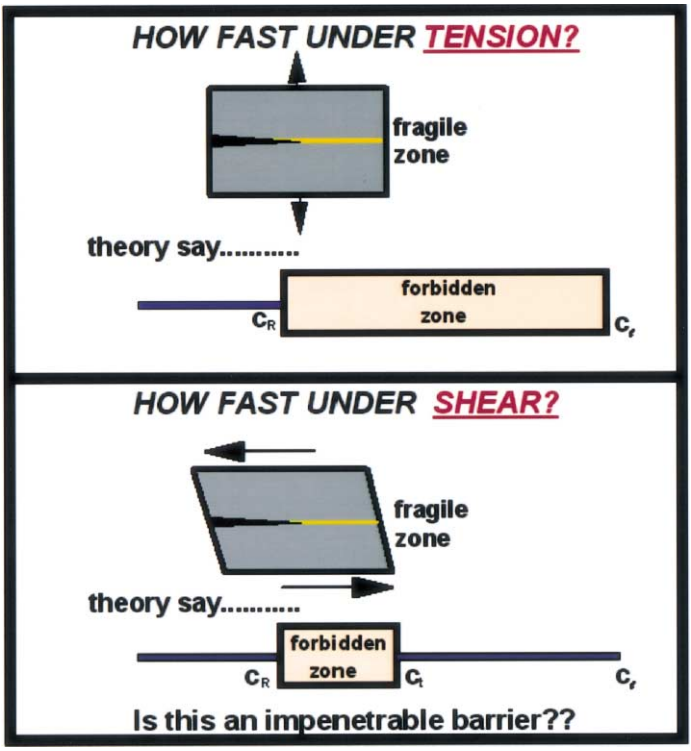


Fig. 2. The forbidden regimes for crack propagation with the different applied loads.

potential is taken to be our simple model potential that allows the atomic bond to break. Fig. 4 shows the dependence of the interatomic potential energy on atomic separation. To study a shear dominated crack, the applied shear strain rate is five times greater than the applied tensile strain rate. The top boundary of the slab is moving up and to the right and the bottom boundary of the slab is moving down and to the left. Linear velocity gradients are established across the slab, and an increasing strain occurs in the solid slab. This leads to eventual failure of the material at the slit tip. The applied strain rates remain constant during the simulation, and the simulation is continued until the growing crack has traversed the total length of the slab. We also conducted simulations of a mode I crack with the same geometrical setup where only an opening strain rate is imposed. The results are discussed below.

The histories of crack velocity under mode I and II loading are easily described. The mode I crack quickly approaches a constant velocity equal to the Rayleigh wave speed of the harmonic crystal showing that the crack velocity is indeed limited by the Rayleigh wave speed and consistent with the classical theories of fracture. The mode II crack also quickly approaches a constant velocity equal to the Rayleigh wave speed. After propagating at this constant velocity for a short while, the crack tip jumps to a higher constant velocity calculated to be the longitudinal sound speed of the harmonic

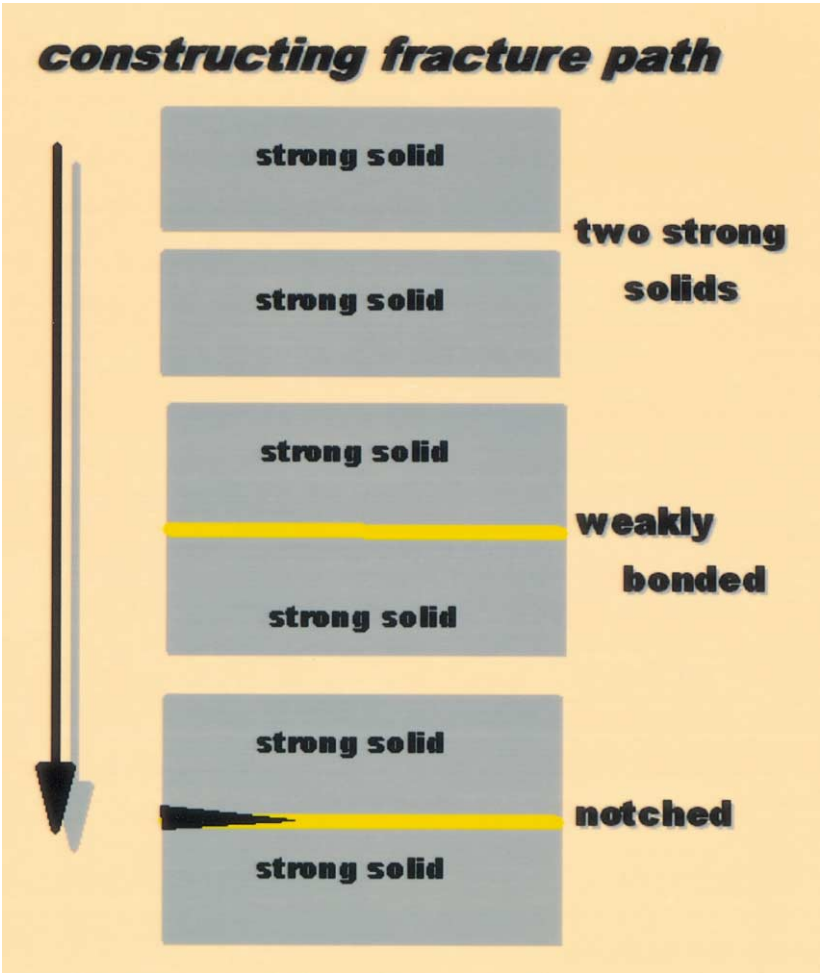


Fig. 3. Construction of the two-dimensional slab that imposes crack propagation along a weak interface joining two strong crystals. The interatomic forces are harmonic except for the fragile zone defined by those pairs of atoms with a separation cutting the vertical centerline of the simulation slab. For these fragile pairs, the interatomic potential is taken to be our simple model potential (the Lennard–Jones potential) that allows the atomic bonds to break.

solid. These results are consistent with the Mach cone angles and the expanding halo configuration shown in Fig. 5. The mode II crack thus exceeded the Rayleigh wave speed and contradicts the classical theories of fracture. Fig. 5 shows several snap shots of a mode II crack accelerating to the longitudinal sound speed. Initially, the mode II crack is subsonic with no shock waves emanating from the crack tip, as shown in Fig. 5(a). As the crack jumps over the forbidden velocity zone and attains the longitudinal wave speed, a pair of shock fronts is observed as Mach cones emanating from the crack tip, as shown in Figs. 5(b) and (c). These shock fronts correspond to

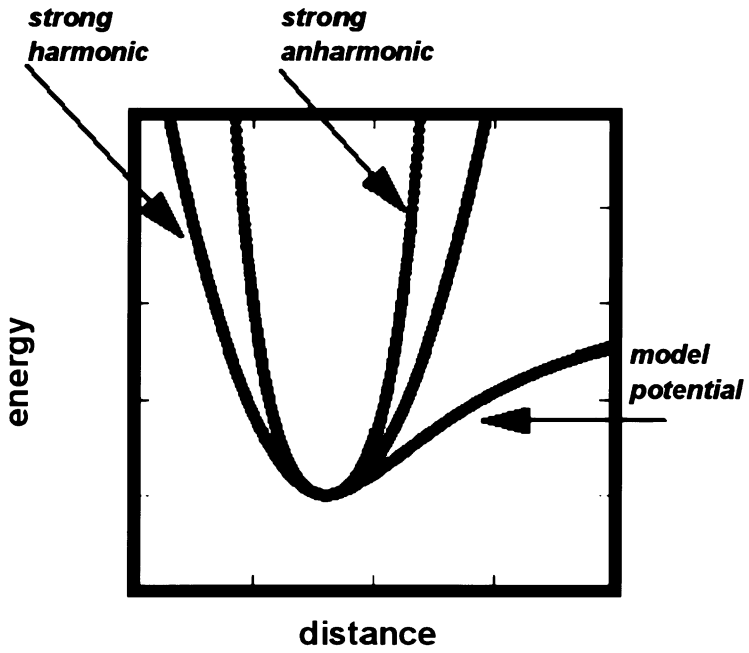


Fig. 4. The dependence of the interatomic potential energy on atomic separation for various force laws used in the simulations discussed in this paper.

transverse sound waves generated at the moving crack. Note that the crack sits at the top of a circular halo that corresponds to longitudinal sound waves emitted from the crack tip. This configuration shows that the crack velocity is equal to the longitudinal wave speed.

The mechanism for the mode II crack "*jumping*" over the forbidden velocity zone is clearly shown in Fig. 6 where a series of color maps of the off-diagonal shear stress component is used to reveal the details of this process. This transition occurs by the nucleation of an intersonic daughter crack ahead of the mother crack traveling at the Rayleigh wave speed. As the mother crack approaches the critical state of nucleation, the crack tip region is asymmetrically distorted with a bulge on the right side of the crack face, as shown in Fig. 5(a). The shear stress distribution near the crack tip is also highly asymmetric. We observe significantly less crack opening at the tip of the daughter crack compared with that of the mother crack, indicating that the daughter crack is more dominantly mode II even though the mother crack has a mode I component. This is consistent with our observation that the mode I crack tends to be limited by the Rayleigh wave speed. Figs. 5(b) and (d) show the detailed process of the birth of the intersonic daughter crack. It is seen that a sharp intersonic crack is nucleated at a small distance ahead of the mother crack. The angle of the Mach cone shows that the velocity of the daughter crack is consistent with the longitudinal wave speed. As the daughter crack moves ahead, the mother crack can still be seen as a surface bulge that trails behind at the Rayleigh wave speed.

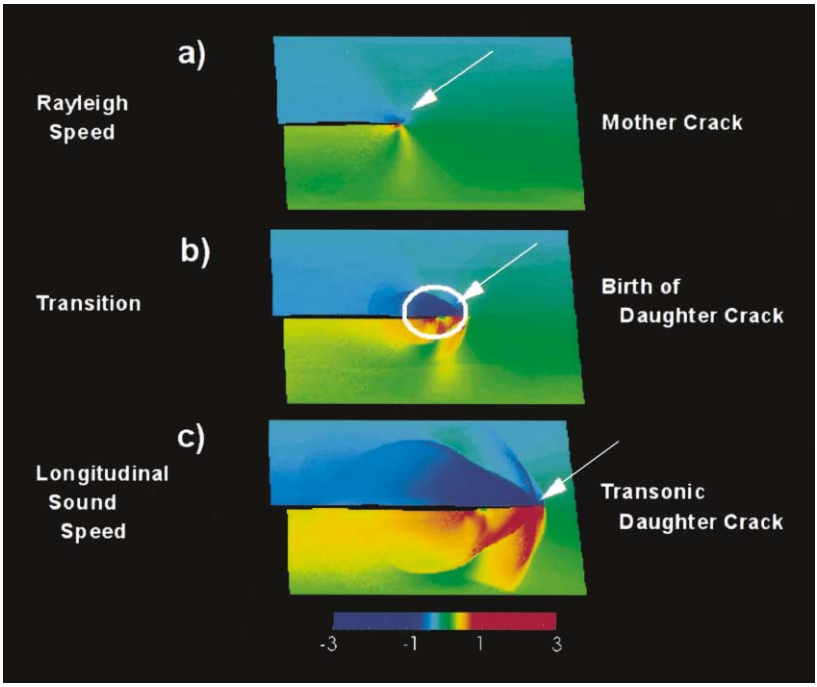


Fig. 5. Transition of a subsonic crack to the longitudinal sound speed. The figures represent a progression in time from the top to bottom. (a) A mode II shear crack travels at the Rayleigh wave speed; (b) The early-time occurrence of shock waves as the crack attains the longitudinal wave speed; (c) The late-time propagating crack traveling at the longitudinal wave speed. The colorbar shows the color map for the local atom speed in the direction perpendicular to the moving crack. The slab dimensions are approximately 800 by 1400 in reduced units, the length scale being the atomic distance when the model interatomic potential crosses over from being repulsive to being attractive.

Our simulations demonstrates intersonic crack propagation and the existence of a "*mother–daughter*" crack mechanism for a subsonic shear crack to jump over the forbidden velocity zone. This mechanism is reminiscent of a similar mechanism based on continuum theories (Burridge, 1973; Andrews, 1976), although the continuum description cannot provide an abinitio description for crack formation and the details of crack tip distortion are not consistent with the continuum solutions. The birth of the daughter crack cannot be characterized by a critical energy release rate or a critical stress intensity factor near the mother crack because both these quantities vanish at the Rayleigh wave speed. It seems that the only possible mechanism by which the daughter crack can be nucleated is by the finite stress peak ahead of the mother crack, and along the weak bonding line, as measured in the stress field and discussed by Burridge (1973). The critical condition for the nucleation of the daughter crack has been investigated using dynamic elasticity equations. The elasticity analysis shows that, while the initiation of the mother crack is governed by the Griffith criterion of fracture, the daughter crack is nucleated by cohesive failure of the weak interface ahead of the

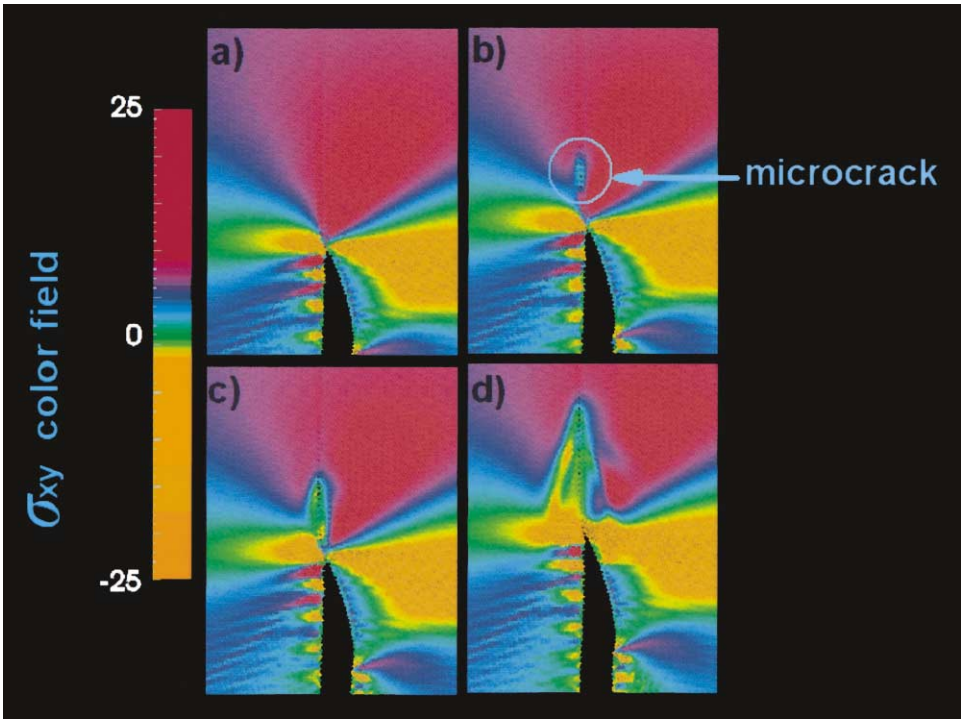


Fig. 6. The nucleation of intersonic daughter crack at the mother crack. The figures represent a progression in time from left to right and from top to bottom which are referenced as (a)–(d). (a) The approach of the critical state for the mother crack. Note the asymmetrically distorted crack tip region; (b) The birth of the intersonic crack. A very sharp slit is born ahead of the tip of the mother crack; (c, d) The daughter crack joins the mother crack and quickly approaches the longitudinal sound speed. The colorbar shows the color map for the shear stress xy component. The bulged mother crack is still propagating at the Rayleigh wave speed. The dimensions of the solid region shown are approximately 90 by 120 (in reduced units).

crack tip. Comparison of the relevant length and time scales predicted by the elasticity analysis with the results from molecular dynamics simulation is excellent (Gao et al., 2001).

We have also made further extensions to this study. For example, we have set the applied strain rates equal to zero at the time of nucleation of the intersonic crack and still observed a constant propagation velocity of the crack at the longitudinal wave speed throughout its passage to complete cleavage. However, the acoustical disturbances are much less explosive. If the slab is allowed to spring back by completely releasing the applied loads as soon as the daughter crack is nucleated, the crack speed drops to the "*square-root of two times the transverse wave speed*" and propagates at this speed until complete cleavage occurs. This loading process seems to closely resemble the dynamic impact loading conditions in the experiments of Rosakis et al. (1999), and our results are entirely consistent with their corresponding experimental observations. The special speed corresponds to the radiation-free speed at which the maximum intersonic

energy release rate occurs. The atomic simulations for transonic crack propagation are discussed in Abraham and Gao (2000).

3.3. Supersonic crack motion

Can cracks go faster than the longitudinal wave speed? The answer is yes. We considered a different interatomic potential for the strong solid. The new spring potential adopted the repulsive part of the model potential both for stretch and for compression. In this case, the solid stiffens with strain, as opposed to the standard model potential for which the solid softens with stretch.

Given the opportunity to be a pioneer user of the new teraflop computer at the San Diego Supercomputer Center, we decided to simulate in three dimensions the crack propagation using this anharmonic spring potential and one billion atoms. This is a new record for total number of atoms simulated by molecular dynamics. The main motivation was to challenge the capabilities of this new supercomputer. It lived up to our expectations. In Figs. 7 and 8, the space-time history of the crack tip is presented. In Fig. 8, snapshot pictures of this 3D crack are shown at different stages of its dynamics corresponding to points along the position–time trajectory of Fig. 7 that are denoted by arrows. We see two shock cones emanating from the moving crack tip. The crack is "*supersonic*" and moving 60% faster than the longitudinal wave

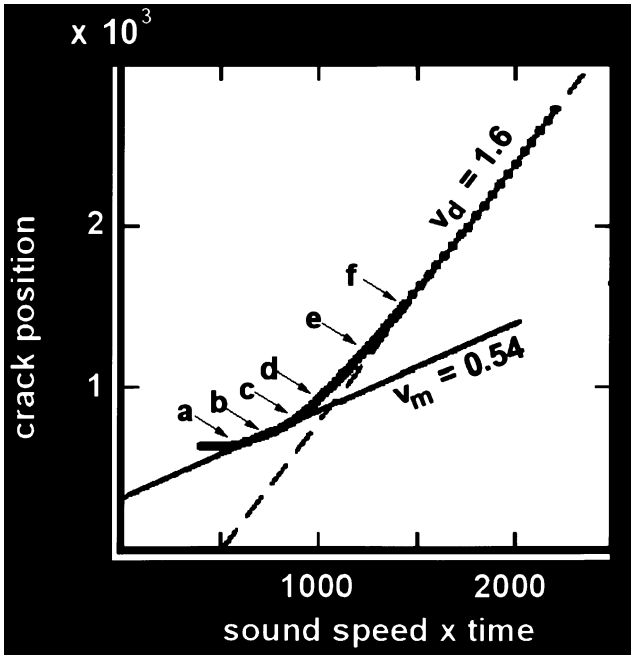


Fig. 7. The space-time history of the crack tip for the three-dimensional, billion-atom simulation of supersonic crack propagation. The final limiting speed of the crack tip is Mach 1.6. The scale of the atomic transverse speed is given by the color bar. Reduced units are used.

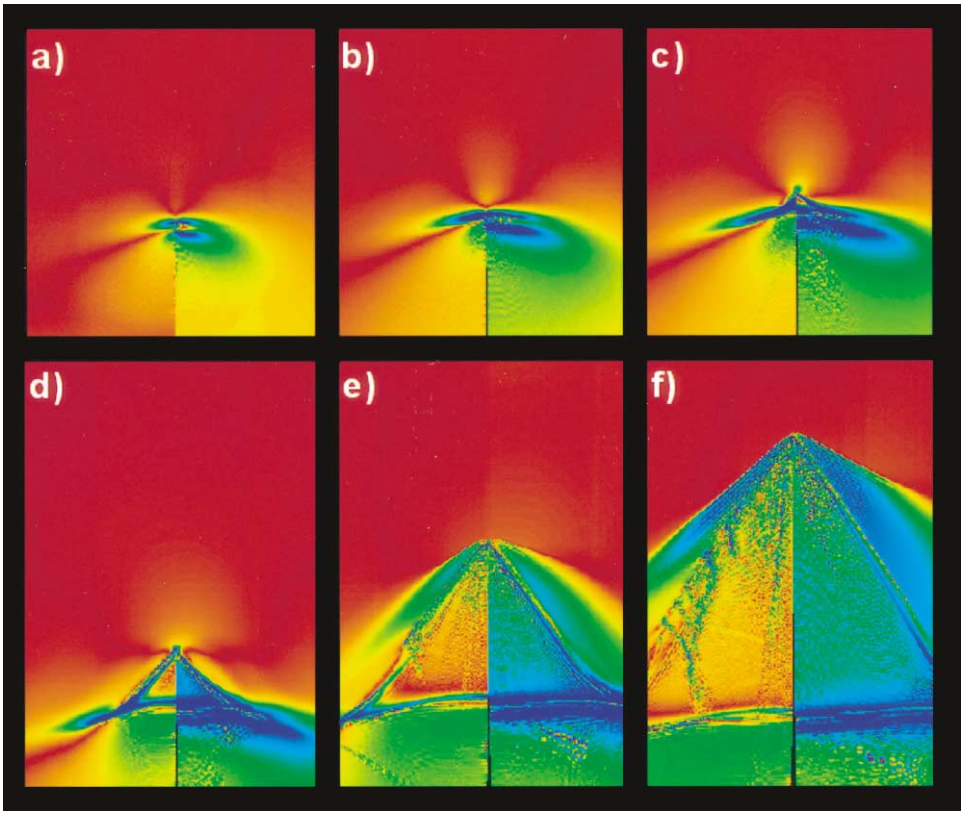


Fig. 8. Snapshot pictures of a 3D crack in a nonlinear solid are shown at different stages of its dynamics corresponding to points along the position-time trajectory that are denoted by arrows in Fig. 7.

speed. This phenomenon is totally contradictory to predictions of classical theories of fracture. The nonlinear properties of the material may provide a possible explanation of this phenomenon. Broberg (1964) and Gao (1993) have shown that such "*hyperelastic*" softening of solids leads to a significant reduction in local wave speeds near a crack tip. We expect that the hyperelastic stiffening of solids will induce significantly higher values of wave speeds near the crack tip. Local wave speeds near the crack tip region have increased significantly beyond the longitudinal wave speed of the harmonic crystal because of the elastic stiffening in this region. As a result, the limiting fracture speed at the crack tip is no longer determined by the harmonic properties of a solid, but rather by its hyperelastic properties. When the elastic disturbance goes beyond the highly deformed tip region, it is traveling supersonically with respect to the harmonic region. Hence, two Mach cones appear.

3.4. A dynamic brittle-to-ductile transition

We have discussed many features of failure associated with rapid brittle fracture. Brittle fracture certainly is not the sole mechanism for materials failure when things

are hit hard and fast; otherwise our world would be quite fragile. Basically, there are two generic classes describing materials failure; a material is brittle or it is ductile. In the first case, chemical bonds are broken, and such a failure is easily recognized when you see glass shatter. For ductile failure, such a catastrophic event does not occur. Tough materials like metals do not shatter; they bend because plastic deformation occurs by the motion of rows of atoms sliding over one another on preferred slip-planes (dislocations) in contrast to bond-breaking. That is why car fenders are not made of brittle materials like glass. These two general classes of failure do not depend on the details of the interatomic interactions, but they do depend on the atomic packing of the three-dimensional solid. Of course, the interatomic potential dictates the packing.

We also recognize that there are a limited number of atomic packings in nature. Glasses do not have extended crystallinity because atoms are packed randomly. They have no slip-planes and therefore cannot exhibit ductility arising from atomic planes slippage. With this picture, glasses should exhibit brittle failure. Three dimensional crystals do not have the isotropy of glasses. In a certain sense, one may view the crystalline solid as a defected solid because of the loss of perfect isotropy, the defect being multiplanar and infinite in extent! Because of the breaking of the perfect isotropic symmetry, slip-planes exist, dislocations are possible, and ductility may win out over brittle failure. The face-centered-cubic (fcc) packing is known to have a strong propensity toward ductility. Crystallinity does not guarantee ductility; strong covalent-bonded solids tend to be brittle.

Our first attempt to go to three-dimensions was a simulation of an fcc solid with a crack and with mode I loading (Abraham, 1997; Abraham et al., 1997a, b). With this atomic packing, one would expect that the solid would eventually undergo ductile failure when the uniaxial stretch became sufficient by crack blunting and no crack extension. On the contrary! The onset of failure is characterized by classic brittle fracture. There is perfect cleavage and crack extension from the accelerating crack up to a point, that point being reached when the crack velocity equals one-third of the Rayleigh wave speed (Fig. 9). Then something quite dramatic happens. The crack stops, and a "*flower of loop dislocations*" blossom from the crack tip. This is shown in Fig. 10. The atomic system is a three-dimensional slab having a total of more than one hundred thousand atoms. We show only those atoms that have a cohesive energy less than 97% of the ideal bulk value, reducing the number of atoms seen by approximately two orders of magnitude. The elastic energy continues to dissipate through the continued creation and motion of the dislocations (Abraham, 1997). Of course, "one-third of the Rayleigh wave speed" is, by now, a very familiar speed in crack dynamics. We conclude that the early brittle fracture dynamics of the fcc solid is an extremely fragile phenomenon and that the onset of the instability nucleates the ductile failure mechanism of crack arrest, dislocation creation and propagation and associated crack blunting. Uniaxial stretching is applied in the [1 1 0] direction for this simulation, and we have argued that it is the hyperelasticity associated with the [1 1 0] direction that allows the initial failure to be brittle (Abraham, 1997). We have observed a "dynamic brittle-to-ductile transition". You might say that we have gone full circle concerning "how fast can cracks propagate".

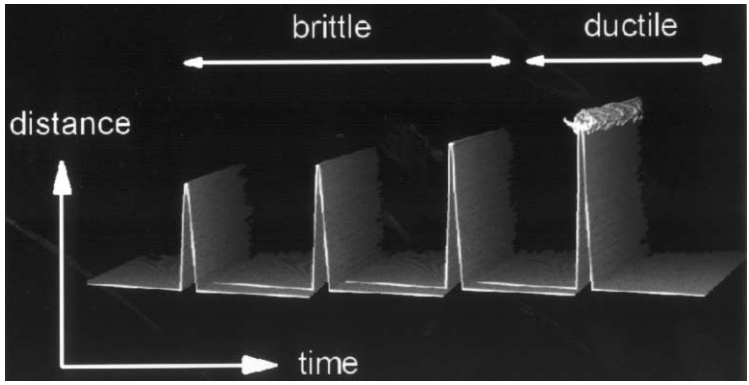


Fig. 9. Early-time sequence of the propagating crack in an fcc solid. Overlapping partial landscapes of the growing crack surface due to brittle fracture are shown in the first three images and the subsequent appearance of dislocations in the fourth image after the transition to ductility. Only atoms with a potential energy less than 97% of the bulk value are displayed, resulting in the selected visualization of atoms neighboring surfaces and dislocations. Approximately a third of the crack's depth is shown.

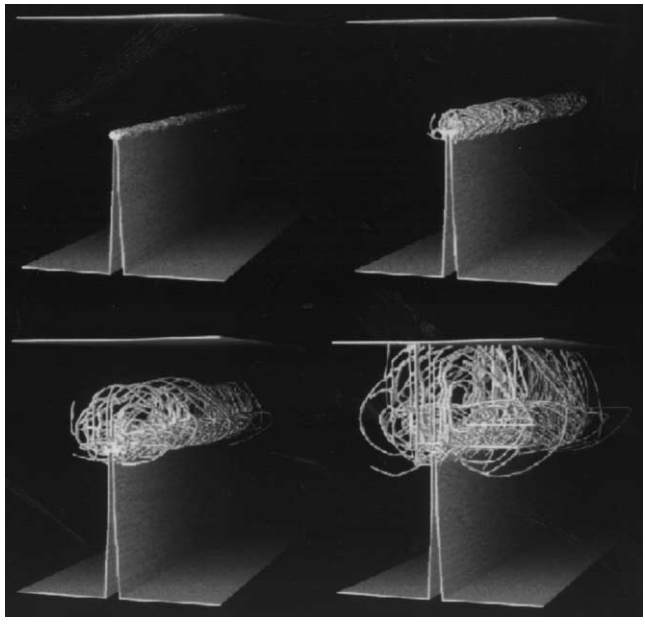


Fig. 10. Late-time sequence of the dislocations propagating through the 100 million atom slab shown for increasing times. With the exceptions that the entire depth and the top surface are shown, the rendering procedure described in Fig. 8 applies.

4. Epilogue

I will discuss new directions for the computer simulation of fracture mechanics. The traditional approach for describing materials phenomena that span a range of length and time scales takes advantage of a hierarchy of physical description. A description of phenomena on a given length scale includes materials specific constants that can be calculated from a theory that describes shorter length scale phenomena; the constants entering this theory can in turn be calculated from a still shorter length scale theory, and so on. For most phenomena of interest to materials science, this sequential coupling of length scales finds its practical end at the level of ab initio theories of electronic structure. The pioneering work of Clementi (1988) provides a lucid example. They used quantum mechanical methods to evaluate the interactions of several water molecules. From this data base, they created an empirical potential for use in atomistic simulations, and evaluated the viscosity of water. Using this computed viscosity, they performed a fluid dynamics simulation to predict the tidal circulation in Buzzard's Bay. This is a powerful example of the sequential coupling of length and time scales: a series of calculations is used as input to the next up the length/time scale hierarchy.

There are many phenomena, however, where physics on short and long length scales interact dynamically; and these cannot be adequately described with a sequential coupling of length scales. Dynamic fracture is a very good example. Energy from macroscopic-scale elastic fields is concentrated on the Angstrom scale of the electrons that participate in atomic bonding. A simulation of this phenomenon requires an accurate description of atoms bonding at the crack tip, while at the same time including very large volumes of strained material. The resolution of information needed on a given scale varies with distance from the crack tip. Far away, it is adequate to use the equations of motion for a macroscopic averaged continuum field. With decreasing distance from the crack tip, singularities in the elastic field are cutoff by atomic scale phenomena and eventually by breaking electronic bonds. These phenomena on the one hand require more information to describe, but on the other hand, are dominant in successively smaller regions of materials. This suggests a natural physical domain decomposition in space: Angstrom cube volumes where electronic excitations must be considered explicitly, nanometer cube regions where atomic processes must be described, and micron cube and greater regions where displacement fields are sufficient. This spatial decomposition makes it possible to combine different simulation methods describing the different physical regions into a single powerful simulation tool.

We have recently presented a method which dynamically couples continuum mechanics far from the crack, empirical-potential molecular dynamics near the crack, and quantum tight-binding dynamics at the crack tip, to simulate fracture in silicon (Abraham et al., 1998a, b). Continuum mechanics has long been fruitfully applied to the study of fracture, by explicitly putting in preexisting cracks or a phenomenological description of material decohesion. We use it to efficiently describe large parts of the system that are elastically deformed but do not include highly strained or broken bonds. Closer to the crack, as the strains become larger and the continuum description becomes less accurate, empirical potential molecular dynamics provides a fast atomistic description. While molecular dynamics can be used to simulate fracture by itself,

the empirical potentials that govern the interaction between atoms may not reliably describe the breaking of bonds at the crack tip. A more accurate description in this region is given by tight-binding quantum mechanics, a method that simulates classical nuclei interacting via a simple quantum-mechanical description of the electrons that form interatomic bonds. We describe the multiscale simulation method as MAAD. The multiscale MAAD approach should be considered only a beginning. Indeed, many improvements are currently being made (Abraham et al., 2000).

Acknowledgements

I acknowledge support from the DOD Maui High Performance Computing Center, the USAF Wright-Patterson Computing Center and the San Diego Supercomputer Center and wish to thank DOD's HPCMO for a Grand Challenge award.

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